

Problems with Neural Networks

Vanishing & Exploding gradients

- weight init
- Activation function
- Batch Normalization
- Gradient Clipping

Not enough data

- transfer learning
- Unsupervised pre-training

Slow training

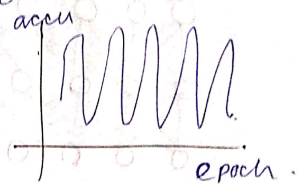
- Optimizers (Adam)
- Learning rate scheduler

Overfitting

- L_1, L_2 Regularization
- Dropouts

While using unscaled data in ANNs →
 there most probably the algorithm cannot converge to the best solution.

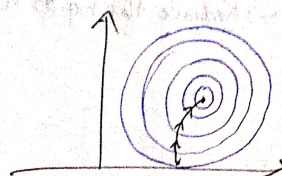
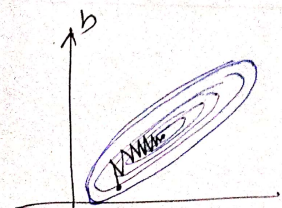
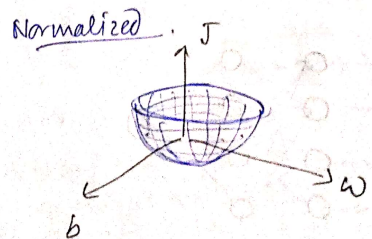
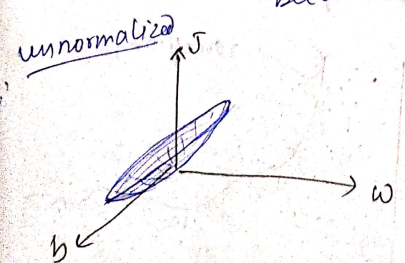
you can see an oscillatory motion in the accuracy curve



age range, range, range
 x_1, x_2 (Salary)

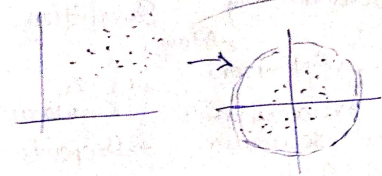
$$w_1, w_2 \rightarrow w_2 = w_1 - \eta \frac{\partial L}{\partial w_1}$$

focus on this weight more because x_2 is more focused.



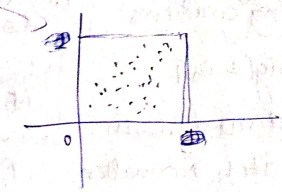
Standardize

$$\frac{x_i - \mu}{\sigma}$$

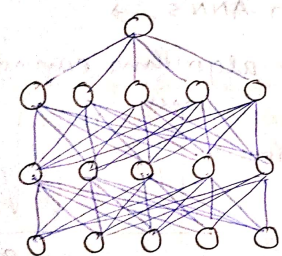


Normalize

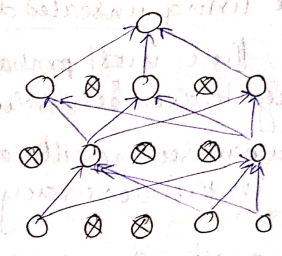
$$\frac{x_i - x_{min}}{x_{max} - x_{min}}$$



The concept of dropouts

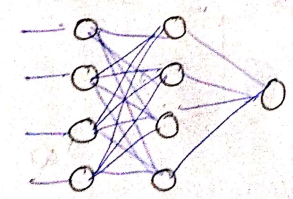


Standard NN

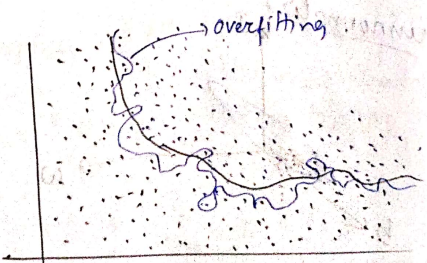


Applying dropout → randomly drop some neurons in each epoch.

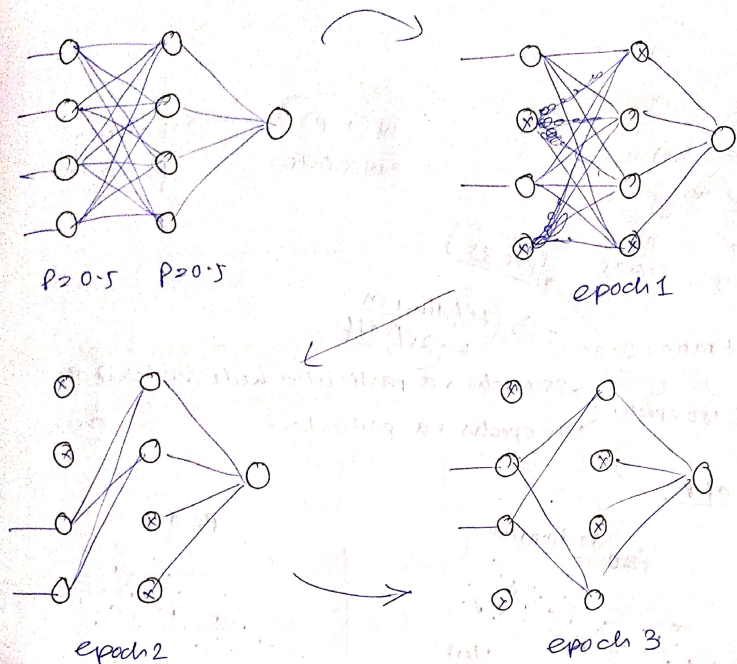
Actually training a fixed data but training with different NNS in each epoch.



→ Reduce the no. of nodes

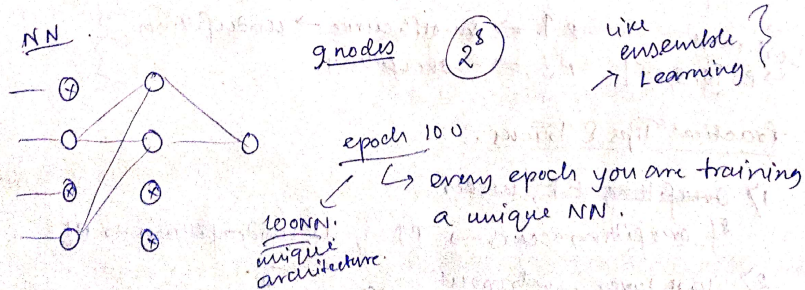
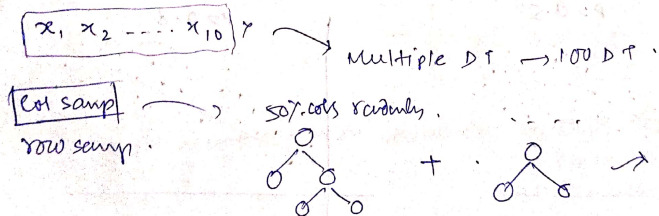


Why this works?

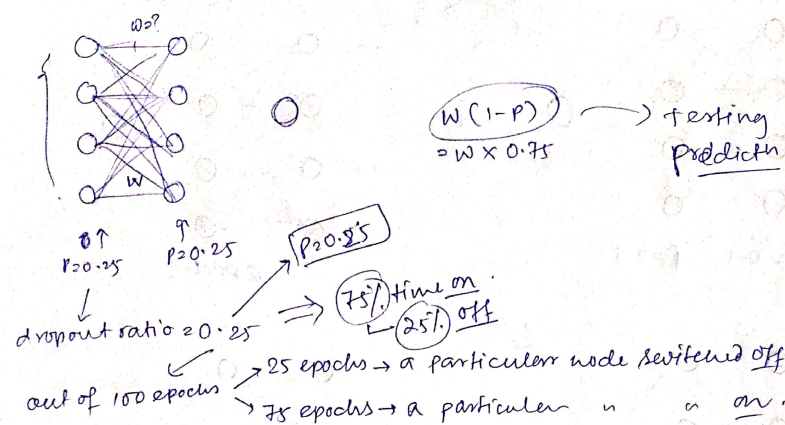


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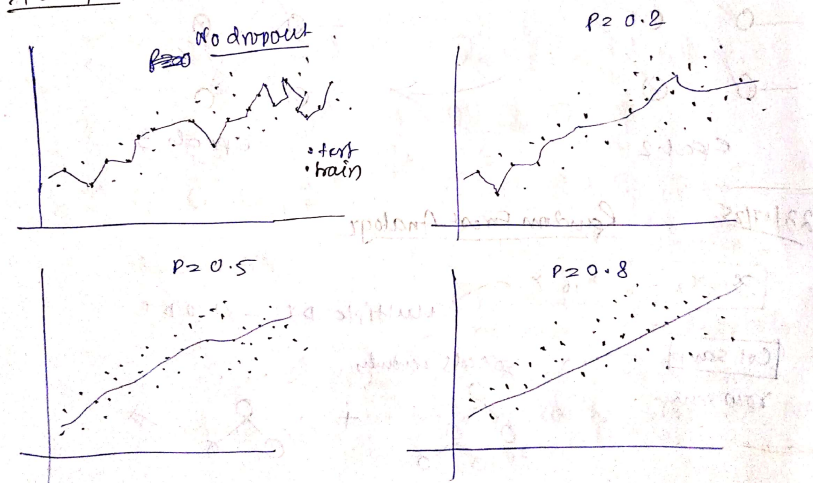
Random Forest Analogy



How Prediction works?



Effect of p .



best range $p \in [0.2, 0.5]$

$p \uparrow \Rightarrow$ smooth curve $\rightarrow$ underfitting

$p \downarrow \Rightarrow$ overfitting

Practical Tips & Tricks.

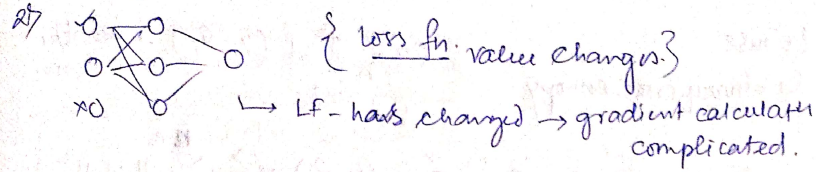
- 1) ~~overfitting~~ $p \uparrow$, under $p \downarrow$ if overfitting occurs $\rightarrow p \uparrow$, if underfitting $\rightarrow p \downarrow$
- 2) last layer $\rightarrow$ dropout.

0	0	0	0
0	0	0	0
0	0	0	0

- 3) CNN $\rightarrow$ 40-50% p
- RNN $\rightarrow$ 20-30% p
- ANN $\rightarrow$ 10-50% p

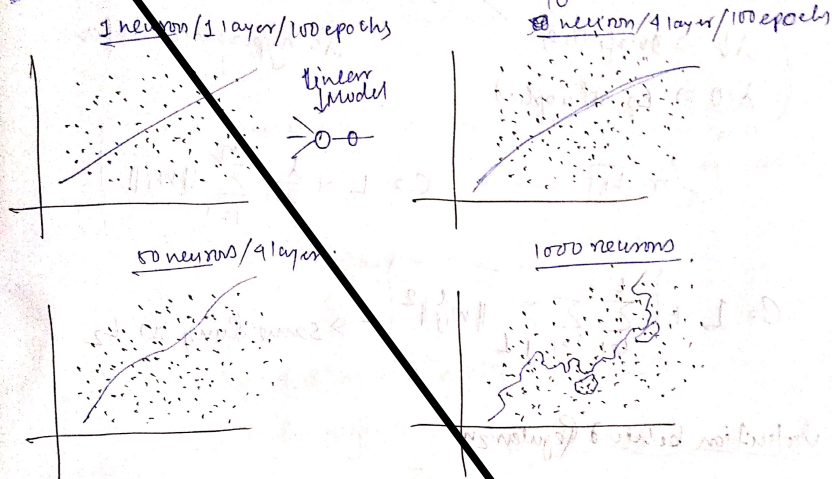
Drawbacks

1 > Convergence $\rightarrow$ delay

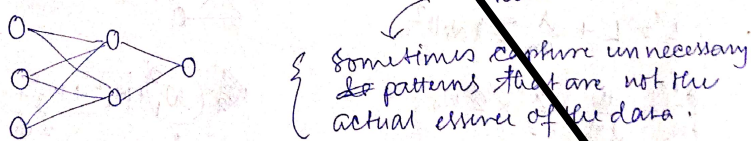


Regularization

Why ANN model overfit?



more neurons $\rightarrow$ more complexity $\rightarrow$ capable to capture minute data from the dataset.



Ways to solve overfitting:

Adding more data

- $\rightarrow$ Add more rows
- $\rightarrow$ Data augmentation

Reducing the complexity of the model

- $\rightarrow$ Dropout
- $\rightarrow$ Early stopping
- $\rightarrow$ Regularization $\rightarrow$ L_1 , L_2 , $L_1 + L_2$

ANN $\rightarrow$ w, b

$\rightarrow$ minimizing loss fn

$L = \text{mse}$

$\hookrightarrow$ binary-cross-entropy

$$C = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{y}_i) + \text{penalty term}$$

[L2] $\rightarrow$

$$C = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{y}_i) + \frac{\lambda}{2n} \sum_{i=1}^n \|w_i\|^2$$

$\lambda \uparrow \Rightarrow$ underfitting $\leftarrow \left(\frac{\lambda}{2n} \right) [w_1^2 + w_2^2 + \dots + w_n^2]$
 $\lambda \downarrow \Rightarrow$ overfitting
 $\lambda = 0 \Rightarrow L_2$ not applied
 λ hyperparameter

for [L1] $\rightarrow C = L + \frac{\lambda}{2n} \sum_{i=1}^n \|w_i\|$

$$C = L + \left[\sum_{l=1}^L \sum_{i=1}^n \sum_{j=1}^n \|w_{ij}^l\|^2 \right] \rightarrow \text{same thing as } L_2$$

Intuition behind Regularization

$$w_1 = w_0 - \eta \frac{\partial L}{\partial w_0}$$

$$L' = L + \frac{\lambda}{2n} \sum \|w_i\|^2$$

$$\frac{\partial L'}{\partial w_0} = \frac{\partial L}{\partial w_0} + \frac{\lambda}{2} \times 2 w_0$$

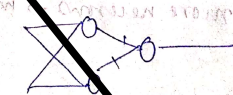
$$= \frac{\partial L}{\partial w_0} + \lambda w_0$$

$$\therefore w_1 = w_0 - \eta \left(\frac{\partial L}{\partial w_0} + \lambda w_0 \right)$$

$$= w_0 - \eta \lambda w_0 - \eta \frac{\partial L}{\partial w_0}$$

$$= (1 - \eta \lambda) w_0 - \eta \frac{\partial L}{\partial w_0}$$

$\downarrow$
 smaller than previous w_0



$$w_1^2 + w_2^2 + \dots + w_n^2$$

$\uparrow$
 differentiate w.r.t. w_1

$L_2 \rightarrow$ Weight decay

$\rightarrow$ In each iteration the value of w gradually decreases.